PROJECT AND TEAM INFORMATION

Project Title

Smart Drone Delivery Management System: Intelligent Drone Allocation, Route Feasibility, and Battery-Aware Scheduling in C++

Student / Team Information

<table>
<tr><td>Team ID: DSCPP-III-2026-T133
Course: B.Tech (CSE), 3rd Semester
Mentor: Dr. Hemant Singh Pokhariya</td><td>Team Name:
DroneMatrix</td></tr>
<tr><td>Team member 1 (Team Lead)
Jaiswal, Suryansh — Roll No: 2028200
Section: C
251001078@geu.ac.in
Role: Overall system flow & finding the best drone logic</td><td></td></tr>
<tr><td>Team member 2
Sahay, Naman — Roll No: 2028776
Section: ML-2
2510370360@geu.ac.in
Role: Battery safety checker & flight distance calculator</td><td></td></tr>
<tr><td>Team member 3
Singh, Nitin — Roll No: 2027950
Section: E
2510013393@geu.ac.in
Role: Order priority queue & live drone tracking maps</td><td></td></tr>
<tr><td>Team member 4
Patel, Hemang Bhai — Roll No: 2027809
Section: J
2510012116@geu.ac.in
Role: City delivery simulation & project testing</td><td></td></tr>
</table>

PROPOSAL DESCRIPTION

Motivation

In the modern world, the delivery industry is faced with serious challenges, including traffic jams, environmental issues, air pollution, and delayed transit of packages. Unmanned delivery drones are able to reach customers straight, however, taking a nearest available drone does not always work well. The drone may deliver a package successfully, but will have no energy left to return to its charging station (Hub → Pickup → Customer → Hub). Wind resistance, changes of elevation, and bigger loads take more energy than expected leading to dangerous crashes. Our solution is to verify the feasibility of a round trip and require at least 15% safety margin for battery before sending the drone on the task.

State of the Art / Current solution

The most common logistics algorithms either use manual human dispatching or basic one-way distance heuristics (Drone → Pickup). Such simplistic approach does not check if the drone is capable of making the trip and returning to its starting point. Moreover, such algorithm is inefficient because of heavy bottlenecks in peak hours, when many orders are coming at once, and the fact that it cannot predict future tasks to assign to the drone that finishes soon.

Project Goals and Milestones

General Goal: To build an intelligent, high-performance, and reliable C++ Drone Delivery Management System using standard data structures to automate fleet allocation, route verification, and battery safety.

Milestone 1 (Phase-I): Formulation of the system architecture, data modeling (Drone, DeliveryRequest, Hub) and multi-factor suitability score formula. **Milestone 2 (Phase-II):** Core C++ data structure implementation (priority_queue, unordered_map, vector), 15% battery safety reserve gate and predictive order queue scheduling. **Milestone 3 (Phase-III):** Urban delivery scenario simulation with 100+ requests, benchmark tests against nearest-first heuristic, and project documentation.

Project Approach

Our C++ delivery management system is divided into four clean, modular components:

- **Delivery Request Manager:** Reads the incoming delivery requests from customer and prioritizes them by urgency and waiting time in min-heap (std::priority_queue).

- **Drone Telemetry Manager:** Monitors the GPS coordinates, battery capacity, and operational state (AVAILABLE, BUSY, CHARGING) of all the drones in hash map (std::unordered_map).

- Feasibility of **Battery** Checkpoint: Ensures that the chosen drone has enough power to undertake the entire journey: Available Battery ≥ Energy(Hub → Pickup → Customer → Hub) + 15% Safety Buffer for Extra Margin.

- **Suitability Scoring** Algorithm: Determines the weighted score S = w1 · dpickup + w2 · ΔE + w3 · twait for each eligible drone, where w1 + w2 + w3 = 1, then assigns the mission to the best drone with the lowest S value.

System Architecture (High Level Diagram)

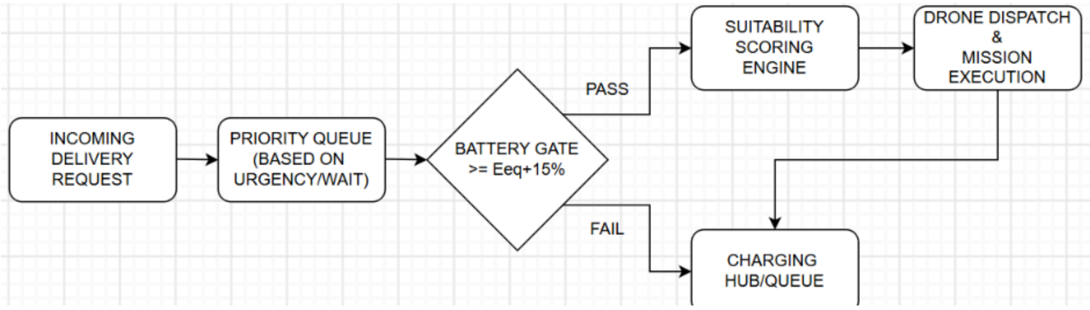

Figure: Complete architectural pipeline showing request ingestion, min-heap priority queueing, safety battery validation, suitability scoring, and automated charging loop.

Project Outcome / Deliverables

- **C++ Fleet Management Engine:** High-performance C++ code using STL data structures (priority_queue, unordered_map, vector) for automatic UAV delivery deployment.
- **City Grid Delivery Simulator: A complete simulation** platform that tests 100+ deliveries on a multi-node urban grid.
- **Benchmark Report:** Empirical benchmarking results showing 0% mid-flight battery depletion incidents and over 35% faster customer delivery time compared to closest first deployment methods.

Assumptions

- Drones fly with a stable nominal speed and consume battery at a consistent rate of usage per kilometer.
- There is an automatic facility where battery can be quickly charged and replaced for exhausted drones.
- Real-time GPS coordinate telemetry is streamed without catastrophic network disconnections.

References

1. Stroustrup, B., *The C++ Programming Language (4th Ed.)*, Addison-Wesley, 2013.
2. Dorling, K. et al., "Vehicle Routing Problems for Drone Delivery," *IEEE Trans. on Systems, Man, and* Cybernetics, 2017.
3. Cormen, T. H. et al., *Introduction to Algorithms (3rd Ed.)*, MIT Press, 2009.